B000311(014)

B.Tech. (Third Semester) Examination
Nov.-Dec. 2022

(Civil Engg. Branch)

MATHEMATICS-III

Time Allowed : Three hours

Maximum Marks : 100

Minimum Pass Marks : 35

Note : *Part (a) is compulsory of each question and attempt any* ***two*** *from remaining.*

Unit-I

1. (a) Evaluate : 4

$$\int_0^\infty t\, e^{-3t} \sin t \, dt$$

(b) Find the inverse Laplace Transform of 8

$$\frac{s+2}{s^2(s+1)(s-2)}$$

(c) Find the inverse Laplace Transform using convolution theorm of 8

$$\frac{s}{(s^2+a^2)^2}$$

(d) Solve : 8

$$t\,y'' + 2y' + t\,y = \cos t, \text{ Given that } y(0) = 1$$

Unit-II

2. (a) Form the partial differential equation from 4

$$z = f(x+at) + g(x-at)$$

(b) Solve : 8

$$(mz - ny)\,p + (nx - lz)\,q = ly - mx$$

(c) Solve : 8

$$r - 4s + 4t = e^{2x+y}$$

(d) Solve 8

$$\left(D^2 + 2DD' + D'^2 - 2D - 2D'\right)z = \sin\left(x + 2y\right)$$

Unit-III

3. (a) Define Expectation and Variance of probability distribution. 4

(b) X is continuous random variable with probability density function given by : 8

$$\begin{aligned} f(x) &= kx, && , (0 \le x < 2) \\ &= 2k && , (2 \le x < 4) \\ &= -kx + 6k , && (4 \le x < 6) \end{aligned}$$

Find k and mean value of X.

(c) The probability that a pen manufactured by a company will be defective is $\frac{1}{10}$. If 12 such pens are manufactured. Find the probability that : 8

(i) Exactly two will be defective

(ii) Atleast two will be defective

(iii) None will be defective

(d) If the probability of a bed reaction from a certain injection is 0·001, determine the chance that out of 2000 individuals more than two will get a bad reaction. 8

Unit-IV

4. (a) Define lagrenge's interpolation formula for unequal intervals. 4

(b) Find the missing term in the table : 8

x	2	3	4	5	6
y	45·0	49·2	54·1	–	67·4

(c) Find the cubic polynomial which takes the following values 8

x	0	1	2	3
$f(x)$	1	2	1	10

Hence or otherwise evaluate $f(4)$.

(d) Give the values : 8

x	5	7	11	13	17
$f(x)$	150	392	1452	2366	5202

Evaluate $f(9)$ using Lagrange's formula.

Unit-V

5. (a) State Taylor series method for solving ordinary differential equation. 4

(b) Obtain a solution upto the fifth approximation of the equation $\frac{dy}{dx} = y + x$, such that $y = 1$ when $x = 0$ using Picard's method. 8

(c) Solve by Taylor's series method the equation $\frac{dy}{dx} = \log(xy)$ for $y(1 \cdot 1)$ and $y(1 \cdot 2)$, given $y(1) = 2 \cdot$ 8

(d) Apply Runge-Kutta fourth order method, to find an approximate value of y when $x = 0 \cdot 2$, given that $\frac{dy}{dx} = y + x$ and $y = 1$ when $x = 0$. 8